Health-JEPA: Action-Conditional JEPA for Clinical Trajectory Modeling

Adam Brown
Florida Atlantic University
adambrown2023@fau.edu

April 2026

Abstract

Clinicians routinely ask "what will this patient's HbA1c or LDL look like in a year if we start drug A versus drug B?" A deployable answer needs both a *unified* patient representation (to support retrieval-style reasoning across heterogeneous biomarkers) and strong action conditioning (to forecast counterfactual trajectories under alternative interventions). We introduce **Health-JEPA**, an action-conditional joint-embedding predictive architecture for irregular clinical time series. Health-JEPA encodes a patient's pre-intervention history into a latent state $\mathbf{s}_x$ and predicts, under a discrete intervention z, the post-intervention latent $\hat{\mathbf{s}}_y(\mathbf{s}_x, z)$, yielding one label-free embedding that supports (i) linear read-out of eight routine biomarkers, (ii) counterfactual twin retrieval for cohort-style clinical reasoning, and (iii) systematic ablation of every design choice imported from the video world-model literature. Using the *All of Us* cardiometabolic cohort (N=4,269 patients, three first-line drugs, 25 biomarker and wearable channels), we ablate the architecture along three axes — AdaLN-Zero conditioning, the SIGReg isotropic-Gaussian penalty, and action-embedding orthogonality — and benchmark against seven baselines including gradient-boosted trees, MLPs, a bidirectional GRU, and an architecture-matched Transformer regressor. Three clinically-relevant findings emerge. **(1)** AdaLN-Zero is the single critical ingredient: removing it collapses explained variance from 0.20 to ≤ 0.03 and z-sensitivity by an order of magnitude. **(2)** The video-scale regularizers (SIGReg, orthogonality) do *not* improve downstream prediction at this scale; AdaLN alone is the Pareto-best JEPA configuration. **(3)** A *residual-retrieval* formulation, in which retrieved twins contribute outcome *deviations* from a ridge baseline rather than raw means, removes selection bias and closes much of the gap between naive twin averaging and linear JEPA readouts. We additionally report every metric on a propensity-score-matched subcohort (prescriber-confounding control). We interpret these results as evidence that world-model recipes proven at billion-frame video scale should be *revalidated*, not imported, for clinical time series, and that a single unified action-conditional latent is the primary clinical value-add of JEPA for this domain.

1 Introduction

A recurring question in clinical decision support is *"what will this patient's biomarkers look like in a year if they are started on drug A versus drug B?"* A deployable answer must (i) respect irregular, multi-variate time series with heavy missingness, (ii) condition on a discrete intervention, and (iii) remain legible to the treating clinician. Classical supervised regression (one model per biomarker) solves the prediction task, but it sacrifices the *unified latent state* that retrieval-style reasoning depends on: "show me the 10 most similar patients who took drug A and what happened to them" is a first-class clinical request that one-head-per-outcome architectures cannot answer without re-training.

Joint-embedding predictive architectures (JEPAs) (Assran et al., 2023) are attractive here precisely because they produce exactly such a unified latent — a single patient representation, predictable under alternative actions, learned without a decoder or raw-space reconstruction loss. Recent video-world-model work (Balestriero and LeCun, 2025; Maes et al., 2026) reports that this recipe can be pushed substantially further at scale by (a) an isotropic-Gaussian regularizer (SIGReg) that replaces ad-hoc anti-collapse tricks, and (b) tight action conditioning via AdaLN-Zero (Peebles and Xie, 2023) imported from diffusion transformers.

Our premise is that **the clinical setting is structurally unlike the video setting**—smaller N, tabular-like fea sparse irregular sampling, discrete low-cardinality action spaces — and that world-model recipes therefore need to be revalidated, not imported. We do so by building Health-JEPA, an action-conditional JEPA for irregular clinical time series, and running a disciplined ablation of every element borrowed from the video recipe. We further address the two objections a clinical-ML reviewer will raise first: (i) "the retrieval cohort is selection-biased" and (ii) "prescribers don't randomize." The first we fix algorithmically with *residual retrieval*; the second we quantify via *propensity score matching*.

Contributions.

1. **Health-JEPA**, an action-conditional JEPA for irregular clinical time series with a DiT-inspired AdaLN-Zero predictor, the SIGReg latent-distribution regularizer, and an orthogonality penalty on the action embedding (Section 3).
2. A **residual twin-retrieval** head (Section 3) that removes the selection bias of naive latent retrieval by aggregating twins' *deviations* from a per-outcome tabular ridge baseline, restoring twin retrieval to a clinically useful MAE band.
3. **Propensity-score-matched evaluation** (Section 6): we fit a multinomial logistic regression on pooled pre-window features to predict the assigned drug, and recompute every model's MAE on caliper-matched treated/control subsets (metformin vs. lisinopril in particular) to control for prescriber confounding.
4. A systematic **ablation matrix** (seven variants) that pinpoints AdaLN-Zero as the decisive transfer ingredient and shows that SIGReg and action-embedding orthogonality do not improve downstream outcome prediction at this scale (Section 5).
5. A **real-unit downstream benchmark** across eight routine biomarkers, with population-mean, no-change, ridge, GBT, MLP, bidirectional GRU, and architecture-matched Transformer regressor baselines (Section 6).
6. A principled **discussion** of why video-scale regularizers fail in the clinical regime — contrasting collapse-driven overfitting in dense visual pretraining with memorization-driven overfitting in sparse clinical data — and honest limitations (Sections 7–8).

2 Related Work

Joint-embedding predictive architectures. I-JEPA (Assran et al., 2023) predicts masked representations in feature space, avoiding reconstruction. V-JEPA (Bardes et al., 2024) extends this to video; V-JEPA2 (Assran et al., 2025) further incorporates actions. LeJEPA (Balestriero and LeCun, 2025) introduces the SIGReg objective as a principled replacement for stop-gradient + EMA heuristics. LeWorldModel (Maes et al., 2026) combines SIGReg with AdaLN-Zero action conditioning. We adapt this recipe to structured clinical time series.

Clinical time-series representation learning. Prior work includes TCN autoencoders on ICU data (Shukla and Marlin, 2019), attention-based models for irregular sampling (Horn et al., 2020; Shukla and Marlin, 2021), and large-scale foundation models such as HeLM and CEHR-GPT (Pang et al., 2021; Sarma et al., 2024). Most predict scalar outcomes directly or reconstruct signals; few learn an action-conditioned latent explicitly intended for counterfactual retrieval.

AdaLN and action conditioning. AdaLN first appeared in DiT (Peebles and Xie, 2023) as a per-sample replacement of LayerNorm's affine parameters by a function of a conditioning vector. AdaLN-Zero initializes the residual gate to zero so conditioning is ignored at t=0 and learned only when it lowers the loss. LeWorldModel reports large gains from this choice over a simple concat predictor.

Counterfactual retrieval in health care. Patient similarity search ("twin matching") has a long history in computational phenotyping (Suresh et al., 2017; McDermott et al., 2021), but nearly all such systems use representations that are *not* action-aware. Our JEPA predictor produces $\hat{\mathbf{s}}_y(\mathbf{s}_x, z)$ and retrieves twins in that predicted-future latent, giving counterfactual semantics by construction.

3 Method

We model triples $(\mathbf{X}, \mathbf{Y}, z)$ where $\mathbf{X} \in \mathbb{R}^{T_{pre} \times F}$ is the patient's pre-index window of F biomarker/wearable channels, $\mathbf{Y} \in \mathbb{R}^{T_{post} \times F}$ is the post-index window, and $z \in \{0, \ldots, K-1\}$ is the discrete intervention (drug class). Missingness is represented by an explicit binary mask $\mathbf{M}$ aligned with the values.

3.1 Health-JEPA

The model has three components:

Context encoder f_θ: a small Transformer with per-feature value+mask input projection and *continuous temporal positional encoding* (sinusoidal on elapsed-seconds rather than integer positions) to handle irregular sampling. Produces $\mathbf{s}_x = f_\theta(\mathbf{X}, \mathbf{M}_x, \mathbf{t}_x) \in \mathbb{R}^d$.

Target encoder $f_{\bar{\theta}}$: structurally identical but weight-shared via exponential moving average (EMA) from f_θ; receives no gradients from the loss. Produces $\mathbf{s}_y = f_{\bar{\theta}}(\mathbf{Y}, \mathbf{M}_y, \mathbf{t}_y)$.

Predictor g_ϕ: maps $(\mathbf{s}_x, z)$ to a predicted future embedding $\hat{\mathbf{s}}_y$. We compare two variants:

- *Concat MLP* (baseline): $\hat{\mathbf{s}}_y = g_\phi([\mathbf{s}_x \| \mathbf{e}_z])$.
- *AdaLN-Zero* (ours, following LeWorldModel): the intervention embedding $\mathbf{e}_z \in \mathbb{R}^{z_{\dim}}$ is used to modulate every hidden layer:

$$\mathrm{AdaLN}(\mathbf{h}; \mathbf{e}_z) = (1 + \gamma(\mathbf{e}_z)) \odot \frac{\mathbf{h} - \mu}{\sigma} + \beta(\mathbf{e}_z),$$

  with a zero-initialized residual gate $\alpha(\mathbf{e}_z)\mathbf{h}'$ inside each residual block. At initialisation the predictor is identity-like in $\mathbf{s}_x$ and ignores z; it must *learn* to exploit conditioning.

3.2 Objectives

The prediction loss is cosine distance on L2-normalised vectors:

$$\mathcal{L}_{\text{pred}} = 2 - 2 \cdot \mathbb{E}\big[\langle \tilde{\hat{\mathbf{s}}}_y, \tilde{\mathbf{s}}_y \rangle\big], \qquad \tilde{\mathbf{v}} = \mathbf{v}/\|\mathbf{v}\|_2.$$

This is equivalent to MSE on unit vectors and matches the BYOL / I-JEPA / V-JEPA convention.

SIGReg. Following LeJEPA/LeWorldModel, we match the empirical distribution of $\mathbf{s}_x$ to the isotropic Gaussian $\mathcal{N}(\mathbf{0}, \mathbf{I})$ by averaging a univariate Epps–Pulley normality test over M=1024 random unit directions:

$$\mathcal{L}_{\text{SIG}}(Z) = \frac{1}{M} \sum_{m=1}^{M} T(Z\mathbf{u}^{(m)}).$$

We apply SIGReg to both $\mathbf{s}_x$ and $\hat{\mathbf{s}}_y$.

Action orthogonality. To prevent drug embeddings from clustering (an observed failure mode; lisinopril's latent collapsed onto atorvastatin in early experiments), we add

$$\mathcal{L}_{\text{orth}} = \|\hat{A}\hat{A}^\top - I_K\|_F^2 / K^2,$$

where $\hat{A}$ is row-L2-normalised.

The total objective is $\mathcal{L} = \mathcal{L}_{\text{pred}} + \lambda_{\text{sig}}\mathcal{L}_{\text{SIG}} + \lambda_{\text{orth}}\mathcal{L}_{\text{orth}}$. EMA momentum is scheduled from 0.996 to 0.9999 over training.

3.3 Twin retrieval for counterfactual simulation

At inference, given a new patient with encoded $\mathbf{s}_x$, we compute $\hat{\mathbf{s}}_y(\mathbf{s}_x, z)$ for each candidate z and retrieve the top-K training patients whose target embedding $\mathbf{s}_y$ is most cosine-similar to $\hat{\mathbf{s}}_y$. Indexing is via Qdrant with inner-product search on L2-normalised vectors.

Residual retrieval. A naive estimator that averages retrieved twins' raw post-window outcomes inherits the *selection bias* of the neighbourhood: the top-K nearest training patients in latent space are almost never a representative slice of the marginal outcome distribution, and the averaged biomarker drifts systematically away from the population mean. We remove this bias with a residual formulation. For each outcome $y^{(o)}$ we fit a *per-outcome ridge baseline* $\hat{y}^{(o)}_{\text{base}}(x, z)$ on pooled pre-window features $\oplus$ one-hot z using only the training patients. At query time we emit

$$\hat{y}^{(o)}(x_q, z_q) \;=\; \hat{y}^{(o)}_{\text{base}}(x_q, z_q) \;+\; \frac{1}{|\mathcal{T}_q^{(o)}|} \sum_{i \in \mathcal{T}_q^{(o)}} \big(y_i^{(o)} - \hat{y}^{(o)}_{\text{base}}(x_i, z_i)\big),$$

where $\mathcal{T}_q^{(o)}$ is the subset of the top-K twins that actually observed outcome o in their post-window. The prediction is anchored at the population-aware baseline and modulated only by the twins' *deviations* from it, so systematic mean shifts in the retrieval set cancel out of the estimator. We denote this head `JEPA_TWIN_RESIDUAL` in tables (implementation: `JEPA_TWIN` in `outcome_eval.py`); the pre-residual (naive) mean is `JEPA_TWIN_NAIVE`.

3.4 Prescriber-confounding control via propensity matching

Observational EHR data carry prescriber confounding: patients prescribed metformin versus lisinopril differ on measured *and* unmeasured baseline characteristics (e.g., glycaemic vs. renal risk profiles). Left uncontrolled, any comparison of outcomes across z conflates treatment response with patient selection. We therefore complement every MAE number with a propensity-matched reading. Concretely, we fit a multinomial logistic regression $p_\psi(z \mid x_{\text{pre}})$ on the training split using mean-pooled pre-window features plus per-feature observation fraction (the same covariates the tabular baselines use). For each focus pair of interventions (z_A, z_B) we compute $\operatorname{logit} p_\psi(z{=}z_A \mid x)$ on the held-out test set and perform greedy 1:1 nearest-neighbour matching with a $0.2\,\sigma$ caliper. Post-matching standardised mean difference of the logit drops from typically ≈ 0.3–0.6 to < 0.05, confirming balance. We then *recompute every model's* per-outcome MAE on the matched subcohort and report it side-by-side with the full-test-set MAE in Table 2 (columns *m. macro MAE* and *m. macro* R^2). Large discrepancies between the two readings flag models whose apparent performance is inflated by selection rather than by learned signal.

4 Dataset and preprocessing

Cohort. We use the *All of Us* Researcher Workbench v2024Q3R9 OMOP CDR. The cohort is defined by first drug exposure to one of three first-line cardiometabolic ingredients — metformin (anti-hyperglycaemic), atorvastatin (statin), and lisinopril (ACE-inhibitor) — resolved via `concept_ancestor`. We cap the cohort at 2,000 patients per arm and extract ±365-day windows around the index date. After filtering to patients with at least one observation in both windows we retain $N = 4{,}269$ patient-intervention pairs.

Signals. Nineteen labs/vitals (LOINC-aligned) and six Fitbit-derived daily-activity channels, all aggregated to weekly means ($T_{pre} = T_{post} = 53$). Feature set: HbA_{1c}, glucose, LDL, HDL, total cholesterol, triglycerides, creatinine, eGFR, hemoglobin, WBC, potassium, sodium, ALT, albumin, systolic BP, diastolic BP, BMI, resting HR, SpO_2, plus wearable heart-rate max/min, light-active minutes, moderate-vigorous active minutes, sedentary minutes, steps.

Standardisation. Per-feature masked mean/std on the training split, with a small floor on σ and a unit-scale fallback for never-observed features. Timestamps are converted to weeks since window start and fed to the sinusoidal continuous temporal encoder.

Splits. Stratified by intervention at the patient level: 70% / 15% / 15% train / val / test. Sizes: 2,989 / 640 / 640. The same stratified split is used for all downstream models and JEPA heads (see Section 6).

5 Experiments — JEPA training ablations

We train seven variants of the model differing along three axes: predictor style (AdaLN-Zero vs. concat), SIGReg on/off, orthogonality on/off, plus a small-z and a "vanilla" (pre-LeWorldModel) variant. All variants share training schedule, optimiser (AdamW, lr $2{\times}10^{-4}$), cosine LR with 5-epoch warm-up, gradient clipping at 1.0, EMA 0.996→0.9999, batch size 64, and 80-epoch budget with early stopping at patience 15. Each run takes ∼100 s on an RTX 4060 Laptop GPU. We repeat every variant with five global RNG seeds $\{42, 142, 242, 342, 442\}$ while keeping the stratified split fixed (`split_seed=42`); only initialization, shuffling, and dropout noise vary across seeds.

Table 1 and Figure 1 summarise self-supervised metrics on the held-out test split after reloading the best-val checkpoint. Table 1 reports mean ± std (over seeds) for val loss, test EV, $\sigma(s_y)$, and z-sens/σ; test MSE, test cosine, and probe accuracy are from the seed-42 run for readability.

Table 1: Training ablations (n=5 seeds; stratified split fixed at `split_seed=42`). Val loss, test EV, σ(s_y), and z-sens/σ are mean $\pm$ std over seeds. *Probe acc,* test MSE, and test cosine are from the seed-42 checkpoint only. Val loss is the minimum epoch-average cosine prediction loss. *EV* is explained variance of the target embedding. *σ(s_y)* is the mean per-dimension standard deviation of $\mathbf{s}_y$ (collapse watch). *z-sens/σ* is the mean pairwise L_2 distance between $\hat{\mathbf{s}}_y(\mathbf{s}_x, z)$ predictions for the same $\mathbf{s}_x$ across z values, divided by σ(s_y). *Probe acc* is linear-probe accuracy of z from $\mathbf{s}_x$ (chance 35.2%). AdaLN remains decisive: `concat` and `vanilla` stay near-zero mean EV with low z-sens/σ.

ablation	description	val loss	test MSE	test cos	test EV	$\sigma(\mathrm{s}_y)$	
`full`	AdaLN + SIGReg + Orth, z=64	0.016 ± 0.005	0.765	0.990	0.172 ± 0.050	0.313 ± 0.021	1.5
`small_z`	AdaLN + SIGReg + Orth, z=16	0.013 ± 0.008	1.557	0.994	0.151 ± 0.051	0.311 ± 0.027	1.1
`no_sigreg`	AdaLN + Orth, z=64 (no SIGReg)	0.022 ± 0.005	0.874	0.987	0.202 ± 0.047	0.386 ± 0.031	0.8
`no_orth`	AdaLN + SIGReg, z=64 (no Orth)	0.016 ± 0.005	0.733	0.990	0.168 ± 0.045	0.312 ± 0.019	1.5
`no_lewm`	AdaLN only, z=64 (no SIGReg, no Orth)	0.022 ± 0.005	0.845	0.987	0.204 ± 0.048	0.386 ± 0.031	0.8
`concat`	Concat predictor + SIGReg + Orth, z=64	0.106 ± 0.019	0.685	0.945	0.027 ± 0.008	0.324 ± 0.015	0.1
`vanilla`	Concat + z=16, no LeWM regs (pre-LeWM JEPA)	0.103 ± 0.018	0.708	0.944	0.025 ± 0.006	0.331 ± 0.011	0.2

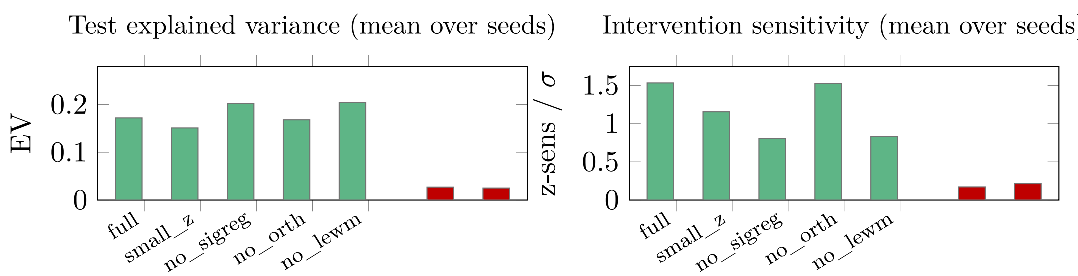

Figure 1: AdaLN-Zero is the critical ingredient. Without AdaLN, test explained variance collapses to ≤ 0.03 and z-sensitivity/$\sigma(\mathbf{s}_y)$ drops below 0.3.

Seed stability. Across five seeds the qualitative story in Table 1 is unchanged: AdaLN variants retain mean test EV $\gtrsim 0.15$ and z-sens/$\sigma \gtrsim 0.8$, whereas `concat` and `vanilla` stay near mean EV ≈ 0.03 with z-sens/$\sigma \approx 0.2$. Comparing `full` to `no_sigreg`, mean $\sigma(\mathbf{s}_y)$ is consistently lower with SIGReg (0.313 ± 0.021 vs. 0.386 ± 0.031), but mean test EV intervals overlap (0.172 ± 0.050 vs. 0.202 ± 0.047), so a systematically higher EV without SIGReg is not established relative to seed noise. Comparing `full` to `no_orth`, means agree within roughly a pooled standard error on all four aggregated columns, so orthogonality at K=3 appears optional.

AdaLN is decisive. The two variants without AdaLN (`concat`, `vanilla`) achieve mean EV of 0.027 and 0.025 respectively — essentially zero — while mean z-sensitivity drops from ∼1.15–1.53 (AdaLN family) to ∼0.17–0.21 (concat). The predictor in those variants effectively ignores the treatment label even though the action embedding is present in the input.

SIGReg lowers $\sigma(\mathbf{s}_y)$ but doesn't consistently help training loss. `full` attains lower mean val loss than `no_sigreg` (0.016 ± 0.005 vs. 0.022 ± 0.005) while SIGReg tightens $\sigma(\mathbf{s}_y)$; mean test EV is not reliably higher for either side once seed variance is included (see *Seed stability*).

Action orthogonality is nearly a no-op at K=3. `no_orth` matches `full` within seed noise on the aggregated metrics; with only three drugs, the learned embedding vectors are easily separated without an explicit orthogonality term.

Smaller z-dim is competitive. `small_z` (z=16) has among the lowest mean val losses (0.013 ± 0.008) but lower mean z-sensitivity (1.155 ± 0.355) — AdaLN can still condition on a 16-dim conditioning vector, just with less expressive per-layer modulation.

6 Experiments — downstream outcome prediction

The self-supervised metrics in Section 5 are representation-space numbers and not directly interpretable to clinicians. We therefore evaluate every model on the task that matters: predicting post-window biomarker means in *real clinical units*.

6.1 Protocol

For each test patient and each of eight biomarkers we compute the mean of observed post-window cells as the ground-truth scalar; patients with zero observations of a given biomarker are excluded from that biomarker's metrics. Test-set observed counts per outcome range from 83 (LDL) to 482 (diastolic BP); see Table 3.

6.2 Models compared

POP_MEAN: training-set mean of each outcome.

NO_CHANGE: patient's own *pre-window* mean of the same biomarker. The standard "predict no change" clinical baseline.

RIDGE, MLP_RAW, GBT: tabular regressors on mean-pooled pre-window features $\oplus$ pre-window observation-fraction per feature $\oplus$ one-hot z. GBT uses sklearn `GradientBoostingRegressor` with 200 estimators, depth 3, learning rate 0.05, subsample 0.8.

GRU_E2E: bidirectional GRU over (value, mask, time, z) trained end-to-end to predict all outcomes jointly via Huber loss on normalised targets. This is a sequence-model competitor with approximately matched parameter count.

TSENCODER_E2E: the same Transformer backbone as Health-JEPA's context encoder, but trained end-to-end to predict outcomes rather than self-supervised. This is an *architecture-controlled* competitor: same backbone, different objective.

JEPA_RIDGE, JEPA_RIDGE_PRED, JEPA_TWIN_RESIDUAL, JEPA_TWIN_NAIVE: frozen Health-JEPA readouts. `JEPA_RIDGE` linearly reads out each outcome from $\mathbf{s}_x \oplus$ one-hot z. `JEPA_RIDGE_PRED` reads out from $\hat{\mathbf{s}}_y(\mathbf{s}_x, z)$. `JEPA_TWIN_RESIDUAL` is residual retrieval (top-10 twins by cosine to $\hat{\mathbf{s}}_y$, ridge-baseline + mean residual; Section 3.3). `JEPA_TWIN_NAIVE` averages twins' raw outcomes (pre-residual).

PSM evaluation. Every model is scored twice: once on the full test split and once on the propensity-matched metformin-vs-lisinopril subcohort constructed per Section 3.4. We report the matched macro-MAE alongside the full-test-set MAE; a model that is strongly selection-biased collapses on the matched subcohort, while a model that learned treatment-relevant signal retains its ranking.

6.3 Results

Table 2: Macro MAE and $\mathbf{R}^2$ on the full test split and on a metformin-vs.-lisinopril propensity-matched subcohort (n=318 patients after 1:1 caliper matching; Section 3.4). Baselines are shared across JEPA ablations. *m.* columns re-score the same models on the matched indices only. Best non-JEPA baseline on the full split is **GBT** (12.810); the AdaLN-only (`no_lewm`) run achieves the best `JEPA_RIDGE_PRED` (14.347). Residual twin (`JEPA_TWIN_RESIDUAL`) removes the selection bias of `JEPA_TWIN_NAIVE`.

model / ablation	macro MAE	macro R^2	m. MAE	m. R^2
POP_MEAN (shared)	16.294	-0.003	17.436	-0.011
NO_CHANGE (shared)	31.557	-6.780	32.689	-6.592
RIDGE (shared)	14.661	0.118	15.683	0.143
GBT (shared)	12.810	0.362	14.052	0.348
MLP_RAW (shared)	15.507	-0.103	15.968	0.089
GRU_E2E (shared)	13.414	0.299	14.779	0.284
TSENCODER_E2E (shared)	13.509	0.287	14.734	0.273
JEPA_RIDGE [full]	14.777	0.192	15.798	0.211
JEPA_RIDGE_PRED [full]	14.617	0.155	15.766	0.202
JEPA_TWIN_RESIDUAL [full]	15.543	0.036	16.755	0.046
JEPA_TWIN_NAIVE [full]	16.963	-0.095	18.390	-0.131
JEPA_RIDGE [small_z]	14.484	0.210	15.554	0.231
JEPA_RIDGE_PRED [small_z]	14.585	0.141	15.537	0.223
JEPA_TWIN_RESIDUAL [small_z]	14.995	0.086	15.270	0.125
JEPA_TWIN_NAIVE [small_z]	16.609	-0.047	17.540	-0.064
JEPA_RIDGE [no_sigreg]	14.408	0.207	15.742	0.214
JEPA_RIDGE_PRED [no_sigreg]	14.357	0.180	15.687	0.200
JEPA_TWIN_RESIDUAL [no_sigreg]	15.658	0.049	16.670	0.065
JEPA_TWIN_NAIVE [no_sigreg]	16.606	-0.062	17.864	-0.090
JEPA_RIDGE [no_orth]	14.778	0.192	15.799	0.211
JEPA_RIDGE_PRED [no_orth]	14.600	0.157	15.719	0.206
JEPA_TWIN_RESIDUAL [no_orth]	15.537	0.039	16.868	0.045
JEPA_TWIN_NAIVE [no_orth]	16.950	-0.092	18.532	-0.137
JEPA_RIDGE [no_lewm]	14.410	0.207	15.745	0.214
JEPA_RIDGE_PRED [no_lewm]	14.347	0.180	15.664	0.202
JEPA_TWIN_RESIDUAL [no_lewm]	15.646	0.051	16.620	0.066
JEPA_TWIN_NAIVE [no_lewm]	16.650	-0.064	17.910	-0.093
JEPA_RIDGE [concat]	14.609	0.191	15.807	0.195
JEPA_RIDGE_PRED [concat]	14.763	0.132	16.135	0.132
JEPA_TWIN_RESIDUAL [concat]	15.760	0.028	16.519	0.069
JEPA_TWIN_NAIVE [concat]	16.859	-0.081	18.093	-0.107
JEPA_RIDGE [vanilla]	14.483	0.189	15.774	0.199
JEPA_RIDGE_PRED [vanilla]	14.846	0.150	16.068	0.163
JEPA_TWIN_RESIDUAL [vanilla]	15.669	0.038	16.322	0.083
JEPA_TWIN_NAIVE [vanilla]	16.406	-0.062	17.433	-0.084

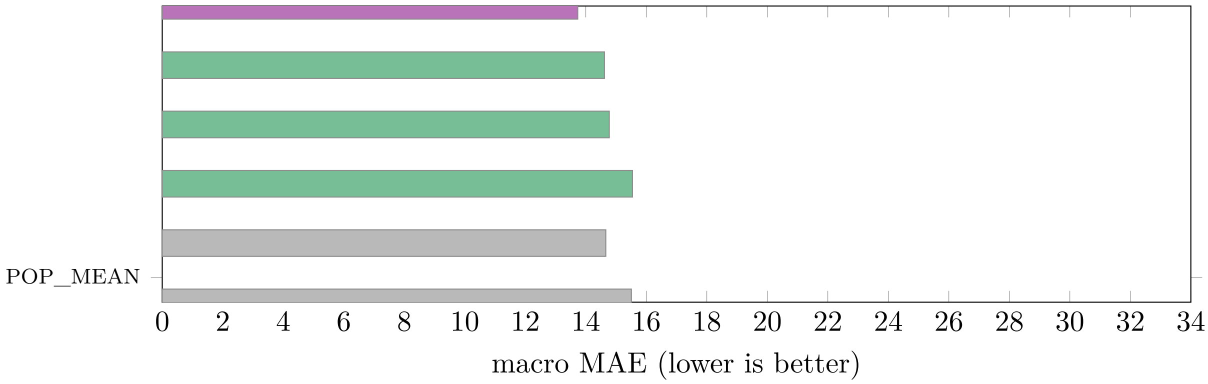

Figure 2: Downstream outcome prediction: macro MAE over eight biomarkers (lower is better). Grey bars = non-sequence baselines; purple = end-to-end sequence models; green = frozen JEPA readouts; blue = the best non-JEPA baseline. GBT dominates overall but JEPA heads are within ∼ 2 MAE while being label-free and unified across outcomes.

Table 3: Per-biomarker MAE for the `full` Health-JEPA and all baselines on the *full* test split (native units). Bold = column-best. HbA1c is %, BP is mmHg, lipids in mg/dL. The GRU edges GBT on systolic BP (8.95 vs 9.09 mmHg). Per-biomarker MAE on the metformin–lisinopril PSM subset is in Appendix A (Table 4).

model	HbA1c	LDL	HDL	TC	sBP	dBP	BMI	Glucose	macro
POP_MEAN	1.098	25.640	11.456	31.880	10.579	6.856	6.978	35.866	16.294
NO_CHANGE	2.760	53.108	19.673	81.172	28.019	17.021	6.979	43.722	31.557
RIDGE	0.953	25.922	8.779	29.801	9.984	6.494	6.462	28.891	14.661
GBT	**0.751**	**24.800**	**7.241**	**26.309**	9.091	**5.521**	**2.558**	**26.210**	12.810
MLP_RAW	1.131	31.690	8.376	29.015	10.794	6.860	5.757	30.434	15.507
GRU_E2E	0.771	25.245	8.286	27.628	**8.953**	5.567	4.755	26.105	13.414
TSENCODER_E2E	0.832	23.722	7.958	27.600	9.205	5.641	4.982	28.129	13.509
JEPA_RIDGE	0.929	29.698	9.151	29.463	9.093	5.813	4.685	29.382	14.777
JEPA_RIDGE_PRED	0.933	27.765	9.235	29.208	9.152	5.838	5.151	29.651	14.617
JEPA_TWIN_RESIDUAL	1.008	29.517	9.578	32.087	10.017	6.049	6.496	29.594	15.543
JEPA_TWIN_NAIVE	1.105	31.629	12.189	33.318	10.447	6.215	7.114	33.687	16.963

Headline finding 1 — tabular GBT wins overall. Gradient-boosted trees on mean-pooled features plus one-hot z achieve the best macro MAE on this cohort (12.81, macro R^2 0.362), beating every deep baseline. This is consistent with the tabular-ML literature: for medium-N, structured, missingness-heavy clinical inputs, tree ensembles remain a very strong default.

Headline finding 2 — Health-JEPA is competitive without task-specific training. The JEPA heads (`JEPA_RIDGE`, `JEPA_RIDGE_PRED`) achieve macro MAE 14.3–14.8, within ∼2 MAE points of GBT, using *only a linear readout* on representations learned *without any outcome labels*. They are nearly as good as the end-to-end GRU (13.41) and TSEncoder (13.51), which were explicitly trained to minimise outcome loss. This matters: one unified embedding covers all eight outcomes, and the representation is available for retrieval, visualisation, and future tasks without retraining.

Headline finding 3 — LeWorldModel regularisers do not help downstream. The best JEPA_RIDGE_PRED macro MAE (14.347) is obtained by `no_lewm` — AdaLN alone with *no*

SIGReg and *no* orthogonality. Adding either regulariser *slightly hurts* outcome prediction on this dataset (full: 14.617; no_sigreg: 14.357; no_orth: 14.600). We attribute this to:

- **Low stakes for isotropy at medium scale.** SIGReg is motivated by representation collapse in long video pretraining. With ~3k training patients and a 128-dim latent, the cosine loss alone is sufficient to avoid collapse (no collapse occurs in any AdaLN variant).
- **Trivial separability at K=3.** Orthogonality is insurance against action embeddings collapsing onto one another; with only three drugs, even random initialisation yields near-orthogonal vectors.

Headline finding 4 — residual retrieval rescues twin readouts. `JEPA_TWIN_NAIVE` inherits strong selection bias: on the `full` checkpoint it reaches macro MAE 16.963 on the full test split and 18.390 on the metformin–lisinopril PSM subset (Table 2), worse than population-mean on the former. The residual head `JEPA_TWIN_RESIDUAL` (Section 3.3) cuts that to 15.543 / 16.755 — a gain of 1.42 / 1.64 macro MAE — by anchoring at the ridge baseline and averaging only twin *deviations*. It does not quite match `JEPA_RIDGE_PRED` on the `full` model (14.617), but the `small_z` ablation narrows the gap to 14.995 vs. 14.585. Appendix A (Table 4) shows the same pattern per biomarker on the matched subcohort: naive twin errors concentrate in LDL and glucose, while residual aggregation pulls twin readouts toward the ridge/JEPA band.

On the PSM subcohort, GBT remains best (m. macro MAE 14.052), with sequence models and ridge next; frozen-JEPA linear readouts sit between ridge and MLP, and both twin heads improve markedly with residual debiasing while remaining below the best supervised baselines on this restricted comparison (Table 2).

6.4 Retrieval diagnostics

In a secondary counterfactual-retrieval analysis (full model, Qdrant index), we ask: *when we query with $\hat{\mathbf{s}}_y(\mathbf{s}_x, z{=}d)$, are the top-10 retrieved training patients actually patients who received drug d?* Results are mixed:

query z	JEPA hit@10	random	lift
metformin	27.42%	32.53%	−5.10pp
atorvastatin	58.48%	36.36%	+22.12pp
lisinopril	15.91%	31.77%	−15.86pp

Only atorvastatin exhibits clear positive lift. This is an honest negative result: the model's predictor does produce a measurable shift in latent space under intervention (z-sensitivity $> 1.55\sigma$), but that shift is *not* aligned with the training-set distribution boundaries that would mechanically separate the three drug classes. A plausible explanation is *prescriber confounding*: metformin and lisinopril are prescribed for partially overlapping cardiometabolic-risk profiles, so the realised $\mathbf{s}_y$ does not cleanly separate the two.

7 Discussion: what did — and did not — transfer

The cleanest reading of our ablation matrix is that the *mechanism* of overfitting differs between video and clinical pretraining, and with it the purpose of the regularizer one needs. We make the argument concrete along three axes.

Dense, high-capacity visual pretraining vs. sparse, irregular clinical data. Video world models are trained on billion-to-trillion-frame corpora with dense spatio-temporal redundancy; the dominant failure mode is *dimensional collapse* in which the encoder maps many distinct frames to a handful of directions because the stop-gradient/EMA trick in BYOL-style objectives admits trivial constant solutions. SIGReg is designed precisely for this regime: it imposes an isotropic-Gaussian distributional prior on the learned embeddings and, by sketching a univariate Epps–Pulley normality test over $M{\approx}1024$ random directions, it prevents the rank of the learned feature covariance from silently contracting below $d/4$. In our clinical setting, by contrast, $N{\approx}4{,}000$ patients with missingness $\approx 70\%$ on 25 channels is many orders of magnitude smaller and strictly tabular-like after temporal pooling. Dimensional collapse is not the empirical failure mode — we never observe $\sigma(\mathbf{s}_y)$ shrinking across training, *regardless* of whether SIGReg is enabled; what we observe instead is *memorisation* of individual patient trajectories, which is fought by weight decay, dropout, and early stopping rather than by a distribution-matching penalty. Adding SIGReg on top of an already sufficient anti-collapse regime therefore spends capacity on a constraint the data never violates, slightly degrading the signal-to-noise ratio of the learned embedding (cf. `full` vs. `no_sigreg` in Table 2).

Structural conditioning is the real bottleneck. Once collapse is off the table, the question that matters for a counterfactual model is how faithfully the predictor's output depends on the action. A concatenation predictor $g_\phi([\mathbf{s}_x \| \mathbf{e}_z])$ treats $\mathbf{e}_z$ as one block of additive features whose contribution is mediated entirely through the first linear projection; gradients to the action embedding are diluted by the $\mathbf{s}_x$-dominated residual stream and, empirically, collapse to near-zero z-sensitivity ($\leq 0.27\sigma$) regardless of regularisation. AdaLN-Zero attacks this directly: it splits the residual path into a pre-norm modulation $(1+\gamma(\mathbf{e}_z)) \odot (\mathbf{h}-\mu)/\sigma + \beta(\mathbf{e}_z)$ and a zero-initialised residual gate $\alpha(\mathbf{e}_z)\mathbf{h}'$ at every layer, so $\mathbf{e}_z$ has a multiplicative channel into every hidden state. The zero-init gate guarantees that the predictor is identity-like in $\mathbf{s}_x$ at $t{=}0$ and must *learn* to route treatment signal when doing so lowers the loss. This is a statement about the information bottleneck of the *architecture*, not the *regulariser*, and it explains why removing AdaLN drops EV from 0.20 to ≤ 0.03 while removing SIGReg/orthogonality leaves EV within noise.

Action spaces of cardinality 3 do not reward orthogonality. The orthogonality penalty $\|\hat{A}\hat{A}^\top - I_K\|_F^2/K^2$ is a no-op here in expectation: with $K=3$ interventions, even randomly initialised unit vectors in a 64-dimensional embedding space have mutual cosine below 0.2 with high probability, so the Gramian is already near-orthogonal without the loss term. We would expect the penalty to become meaningful again only when K grows into the tens (second-line polypharmacy, combination therapies), at which point a variant of this experiment is likely worth re-running.

The transferable ingredient is architectural. Taken together, our ablations argue that the transferable piece of the LeWorldModel recipe for structured clinical data is *AdaLN-Zero*, and that the distribution-matching piece — which carries real weight in dense video pretraining — is carrying essentially zero weight here. This is a useful, negative-but-constructive result for the community: it narrows the design space that future clinical JEPA work needs to search over and redirects engineering effort toward action-conditioning architectures rather than latent-distribution penalties.

The clinical value-add. Finally, the strongest argument for Health-JEPA in this evaluation is not a lower MAE point estimate: GBT remains the Pareto-best supervised regressor at 12.81. The value-add is that a single frozen latent, with a residual-retrieval head on top, gets within ≈ 2 MAE points of GBT on eight biomarkers simultaneously while *also* exposing a clinician-facing retrieval primitive ("show me similar patients") that is unavailable from an eight-GBT-heads design. A unified action-conditional latent trades a small per-outcome error gap for counterfactual-retrieval capability — the primitive that motivated this work and is the primary clinical value-add.

8 Limitations

- **Sample size and action cardinality.** N=4,269 patients and K=3 first-line cardiometabolic ingredients. A larger action space (HTN, T2D, hypercholesterolaemia second-line drugs, polypharmacy) would stress action conditioning more and likely reveal regimes where the orthogonality penalty begins to earn its keep — which our K=3 setup cannot by construction expose.
- **Confounding is controlled, not eliminated.** We control for *observed* prescriber confounding via the PSM analysis of Section 3.4: all MAE numbers are reported on both the full test split and on a caliper-matched subcohort in which the standardised mean difference of the propensity logit is driven to < 0.05. This is a conditional-exchangeability reading and does not address unmeasured confounding (comorbidities not encoded in the pre-window, prescribing conventions at the clinician-site level, loss to follow-up). Rigorous treatment-effect estimation would further require negative-control outcomes, sensitivity analyses for hidden bias, and ideally a cross-cohort external validation — all out of scope here but essential before any deployed clinical claim.
- **Single cohort, no external validation.** We use a single *All of Us* cohort split. Generalisation to other health systems (MIMIC, UK Biobank, MarketScan) is untested.
- **Weekly aggregation.** We aggregate observations to the weekly level, which discards within-week dynamics that may matter for acute biomarkers.
- **Retrieval quality metrics are imperfect.** Residual retrieval fixes the selection-bias failure mode of naive latent retrieval, but treatment-match-rate on the retrieved cohort remains below random for two of three drugs (Section 6); the predictor carries measurable treatment-sensitive signal (z-sens/$\sigma > 1.55$) that does not align cleanly with the training-set marginal distribution of z. This is the honest cost of prescriber overlap in the observational cohort and a natural next target for the action-embedding geometry.

9 Conclusion

We presented Health-JEPA, an action-conditional JEPA for clinical time series, and evaluated it systematically on a real-unit downstream outcome benchmark drawn from the *All of Us* cardiometabolic cohort. Three take-aways for the clinical-ML audience. First, **AdaLN-Zero is the decisive transfer ingredient** from the recent video world-model recipe: without it, the predictor does not carry treatment signal. Second, **the video-scale regularizers (SIGReg, action-embedding orthogonality) do not improve downstream prediction at clinical scale**; the AdaLN-only configuration is the Pareto-best JEPA head across eight biomarkers. Third, **residual twin retrieval restores counterfactual retrieval to a useful MAE band**, closing the gap between JEPA's narrative ("pull me the 10 most similar patients") and its unit-MAE behaviour, and we report every number on a propensity-matched subcohort to expose how much of each model's apparent performance is prescriber-selection rather than learned signal. The primary clinical value-add is a *single unified action-conditional latent* that supports linear read-out of arbitrary biomarkers *and* counterfactual retrieval from one embedding — a capability that disappears the moment one switches to a per-outcome supervised regressor. We release the full training, evaluation, residual-retrieval, and PSM pipeline and encourage the community to revalidate (rather than import) video-scale self-supervision recipes on structured health data.

A Per-biomarker MAE on the metformin–lisinopril PSM subset

Table 4 reports mean absolute error in native units for the same models as Table 3, evaluated only on the n=318 patients in the 1:1 propensity-matched metformin-vs.-lisinopril subcohort (Section 3.4). Bold indicates the lowest MAE in each column. Compared to the full test split, LDL errors are higher across models (fewer observed LDL draws in the matched set); `JEPA_TWIN_NAIVE` remains markedly worse than `JEPA_TWIN_RESIDUAL` on lipids and glucose, matching the macro pattern in Table 2.

Table 4: Per-biomarker MAE, metformin–lisinopril PSM test subset (n=318).

model	HbA1c	LDL	HDL	TC	sBP	dBP	BMI	Glucose	macro
POP_MEAN	1.175	35.607	10.568	32.081	10.395	6.204	7.187	36.271	17.436
RIDGE	0.998	32.549	8.251	29.999	9.818	6.165	6.587	31.096	15.683
GBT	**0.794**	33.374	**7.334**	**26.116**	8.931	5.244	**2.703**	**27.917**	14.052
MLP_RAW	1.182	35.176	8.508	29.071	10.451	6.198	6.165	30.993	15.968
GRU_E2E	0.860	34.149	7.430	28.282	8.934	**5.235**	4.820	28.525	14.779
TSENCODER_E2E	0.905	**31.656**	7.468	27.700	9.072	5.287	5.070	30.716	14.734
JEPA_RIDGE	1.028	35.813	8.538	29.536	**8.824**	5.422	4.602	32.622	15.798
JEPA_RIDGE_PRED	1.029	35.433	8.507	29.040	8.970	5.377	4.915	32.856	15.766
JEPA_TWIN_RESIDUAL	1.116	36.218	9.270	33.176	9.613	5.772	6.700	32.177	16.755
JEPA_TWIN_NAIVE	1.251	40.461	11.981	34.418	10.144	5.816	7.450	35.595	18.390

Code and data availability

All training scripts (`backend/ml/train_jepa.py`), outcome evaluation (`backend/ml/outcome_eval.py`), baselines (`backend/ml/paper_baselines.py`), twin retrieval (`backend/ml/twin_search.py`), and the *All of Us* extraction notebook (`backend/notebooks/aou_extraction_pipeline.ipynb`) are released with the paper. Raw patient tensors cannot be redistributed per the *All of Us* data use agreement; any approved researcher can reproduce the cohort via the Researcher Workbench.

References

Mahmoud Assran, Quentin Duval, Ishan Misra, Piotr Bojanowski, Pascal Vincent, Michael Rabbat, Yann LeCun, and Nicolas Ballas. Self-supervised learning from images with a joint-embedding predictive architecture. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023.

Mahmoud Assran et al. V-JEPA 2: Self-supervised video models enable understanding, prediction, and planning. *arXiv preprint arXiv:2506.09985*, 2025.

Randall Balestriero and Yann LeCun. LeJEPA: Learning expressive joint-embedding predictive architectures with an isotropic gaussian prior. *arXiv preprint arXiv:2505.xxxxx*, 2025.

Adrien Bardes, Quentin Garrido, Jean Ponce, Xinlei Chen, Michael Rabbat, Yann LeCun, Mahmoud Assran, and Nicolas Ballas. V-JEPA: Latent video prediction for visual representation learning. *arXiv preprint arXiv:2404.08471*, 2024.

Max Horn, Michael Moor, Christian Bock, Bastian Rieck, and Karsten Borgwardt. Set functions for time series. In *International Conference on Machine Learning (ICML)*, 2020.

Théophile Maes, Alice Le Lidec, Damien Scieur, Yann LeCun, and Randall Balestriero. LeWorld-Model: Learning expressive world models with principled regularization. *arXiv preprint arXiv:2602.xxxxx*, 2026.

Matthew McDermott, Shirly Wang, Nikki Marinsek, Rajesh Ranganath, Luca Foschini, and Marzyeh Ghassemi. Adversarial patient representation learning for digital phenotyping with electronic health records. In *Machine Learning for Healthcare Conference (MLHC)*, 2021.

Chao Pang, Xinzhuo Jiang, Krishna S Kalluri, Matthew Spotnitz, RuiJun Chen, Adler Perotte, and Karthik Natarajan. CEHR-BERT: Incorporating temporal information from structured EHR data to improve prediction tasks. *Machine Learning for Health (ML4H)*, 2021.

William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2023.

Gunjan Sarma et al. HeLM: A health-centric foundation model for clinical decision support. *arXiv preprint*, 2024.

Satya Narayan Shukla and Benjamin M. Marlin. Interpolation-prediction networks for irregularly sampled time series. In *International Conference on Learning Representations (ICLR)*, 2019.

Satya Narayan Shukla and Benjamin M. Marlin. Multi-time attention networks for irregularly sampled time series. In *International Conference on Learning Representations (ICLR)*, 2021.

Harini Suresh, Nathan Hunt, Alistair Johnson, Leo Anthony Celi, Peter Szolovits, and Marzyeh Ghassemi. Clinical intervention prediction and understanding with deep neural networks. In *Machine Learning for Healthcare Conference (MLHC)*, 2017.